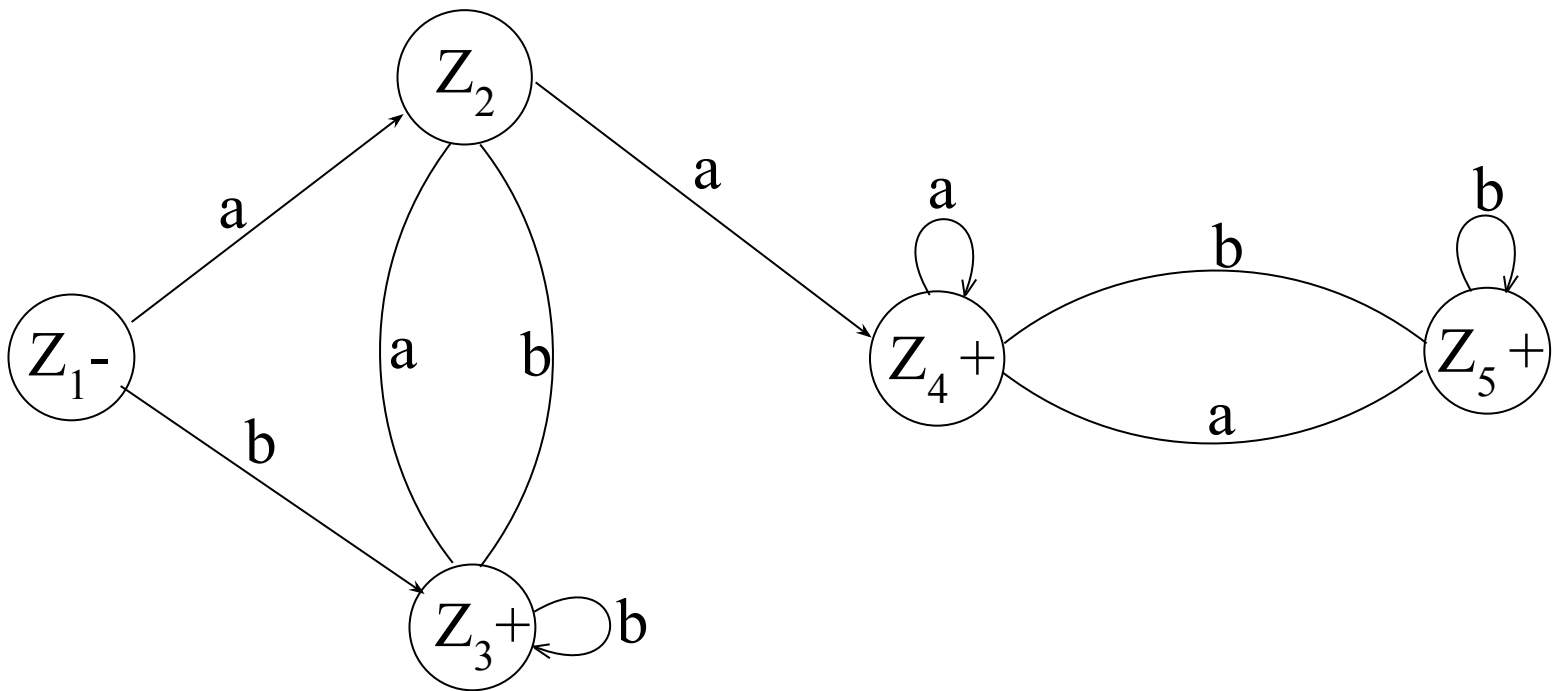


RECAP Lecture 13

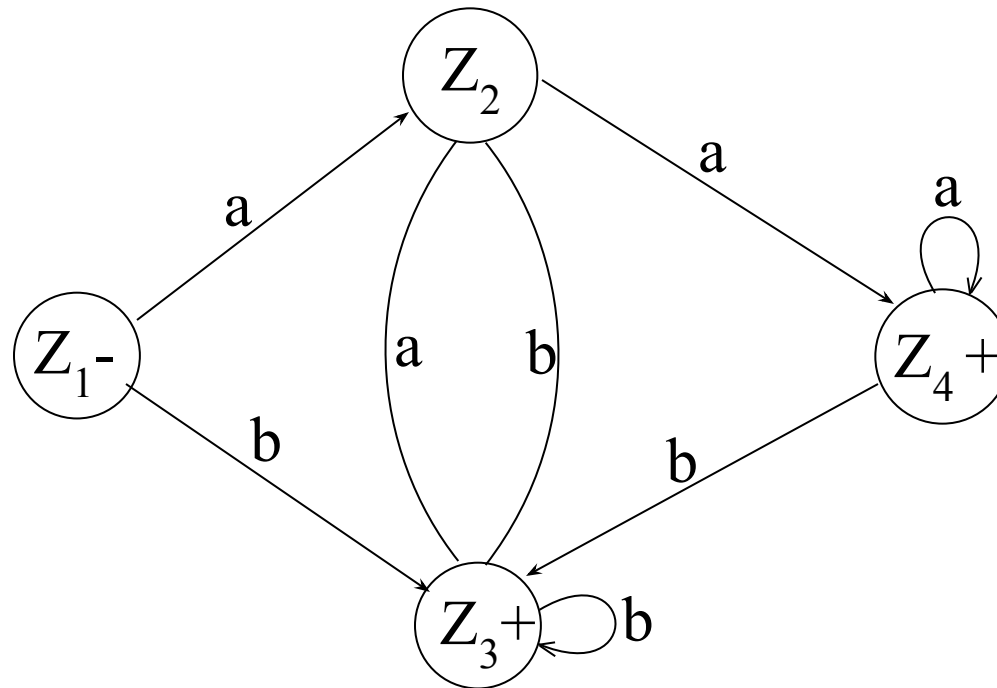
- Examples of Kleene's theorem part III
(method 1) continued ,Kleene's theorem part III
(method 2: Concatenation of FAs),
- Example of Kleene's theorem part III
(method 2 : Concatenation of FAs)

Task

- Build an FA equivalent to the following FA

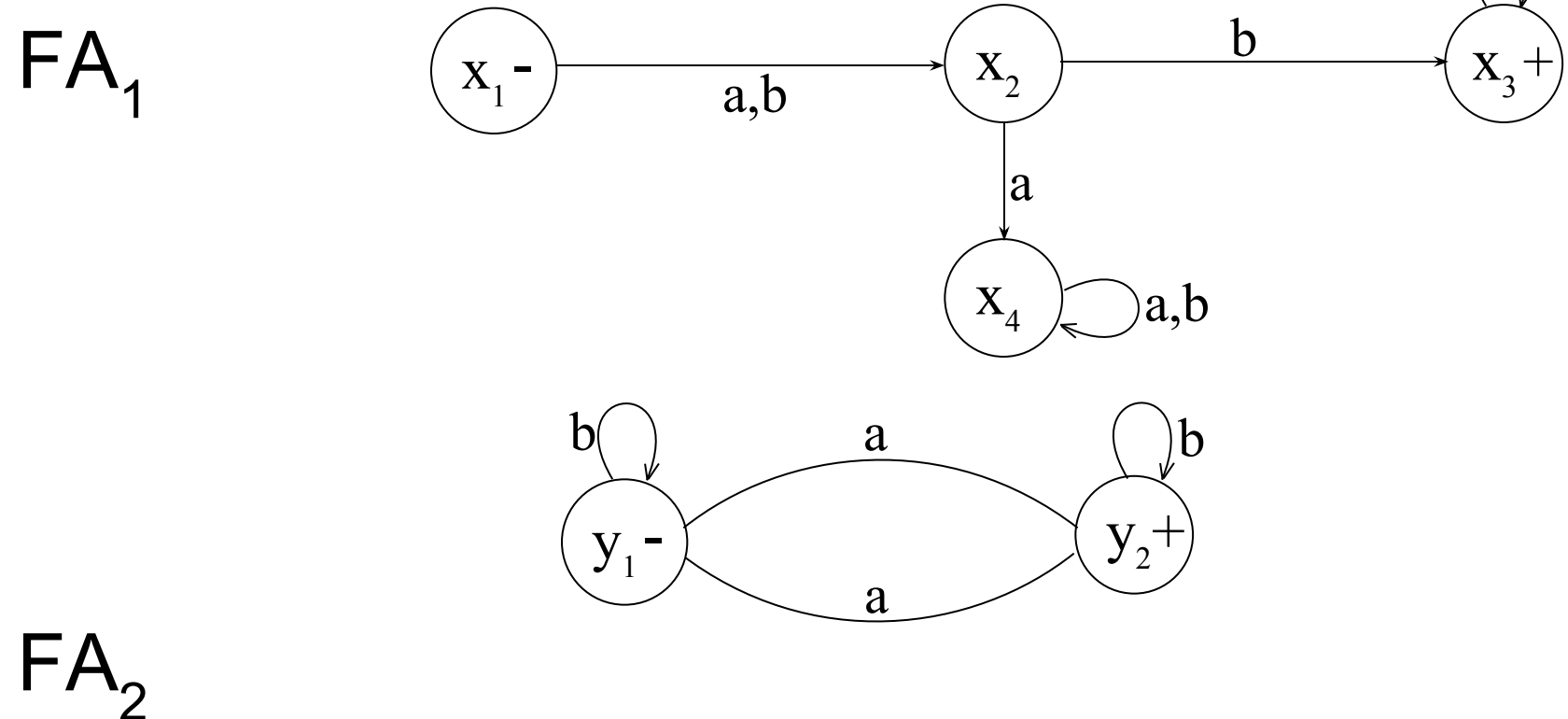


Solution of the Task



Task

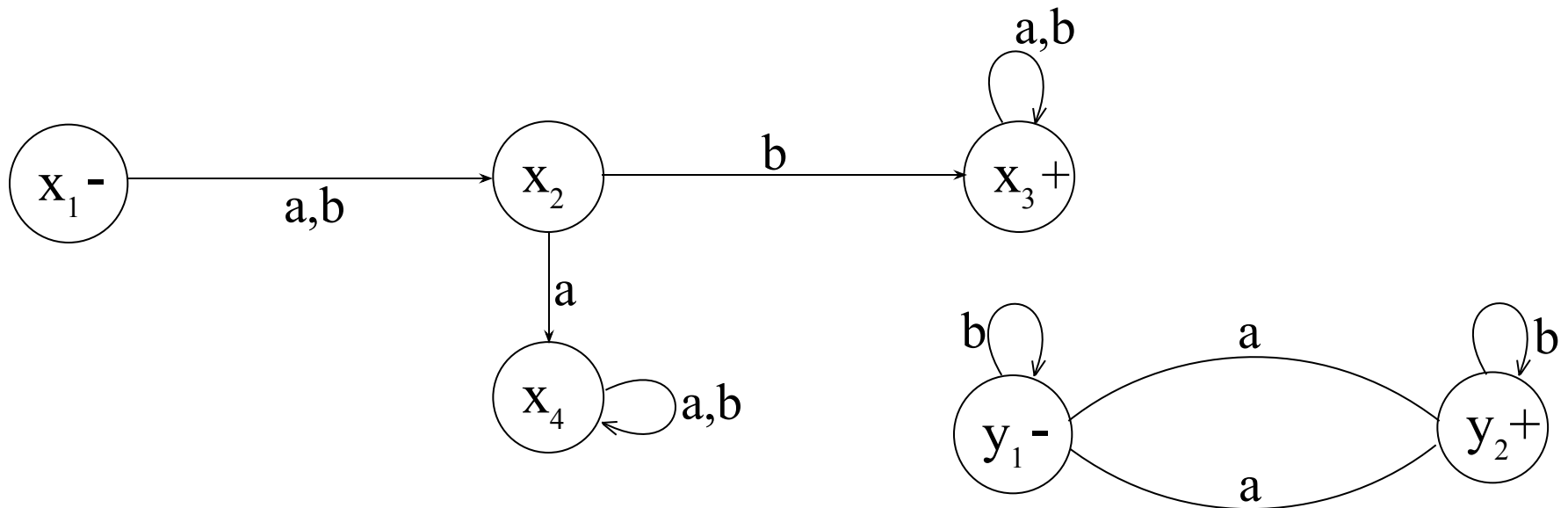
Build an FA corresponding to the union of these two FAs *i.e.* $FA_1 \cup FA_2$ where



Task solution

- RE corresponding to FA_1 may be $(a+b)b(a+b)^*$ which generates the language of strings, defined over $\Sigma=\{a,b\}$, **with b as second letter.**
- RE corresponding to FA_2 may be $b^*a(b+ab^*a)^*$ which generates the language of strings, defined over $\Sigma=\{a,b\}$, **with odd number of a's.**

Solution continued ...

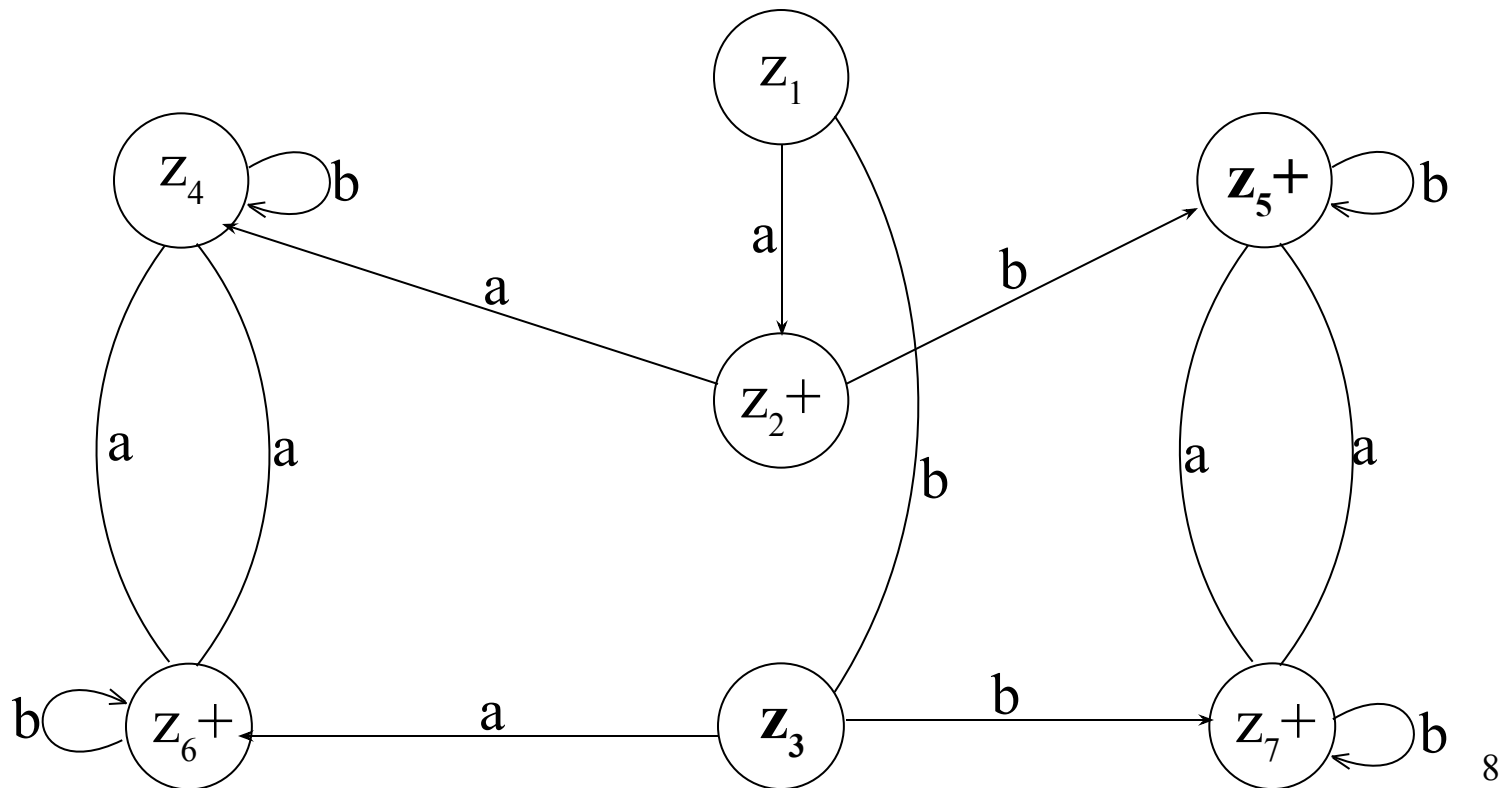


Old States	New States after reading	
	a	b
$z_1^- \sqcap (x_1, y_1)$	$(x_2, y_2) \sqcap z_2$	$(x_2, y_1) \sqcap z_3$

Solution continued ...

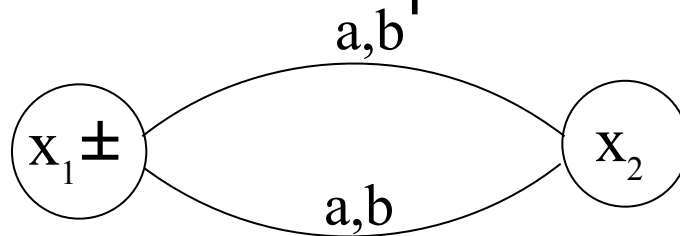
Old States	New States after reading	
	a	b
$z_2^+ \sqcap (x_2, y_2)$	$(x_4, y_1) \sqcap z_4$	$(x_3, y_2) \sqcap z_5$
$z_3 \sqcap (x_2, y_1)$	$(x_4, y_2) \sqcap z_6$	$(x_3, y_1) \sqcap z_7$
$z_4 \sqcap (x_4, y_1)$	$(x_4, y_2) \sqcap z_6$	$(x_4, y_1) \sqcap z_4$
$z_5^+ \sqcap (x_3, y_2)$	$(x_3, y_1) \sqcap z_7$	$(x_3, y_2) \sqcap z_5$
$z_6 \neq \sqcap (x_3, y_1)$	$(x_3, y_2) \sqcap z_5$	$(x_4, y_1) \sqcap z_4$

Solution continued ...

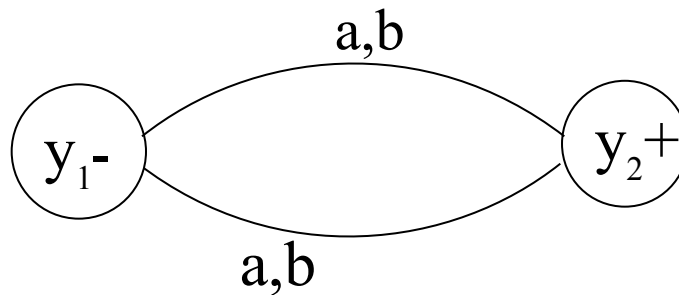


Example

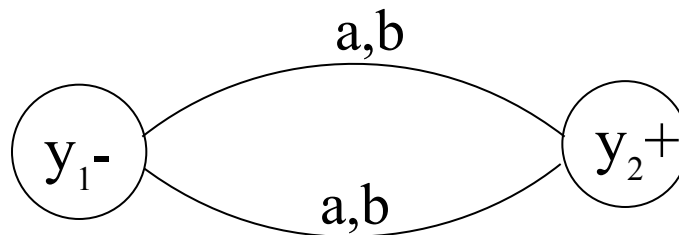
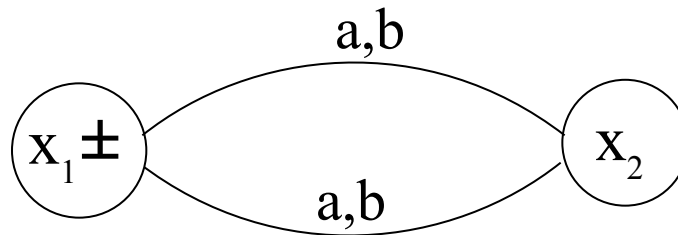
Let $r_1 = ((a+b)(a+b))^*$ and the corresponding FA_1 be



also $r_2 = (a+b)((a+b)(a+b))^*$ or $((a+b)(a+b))^*(a+b)$ and FA_2 be



Example continued ...

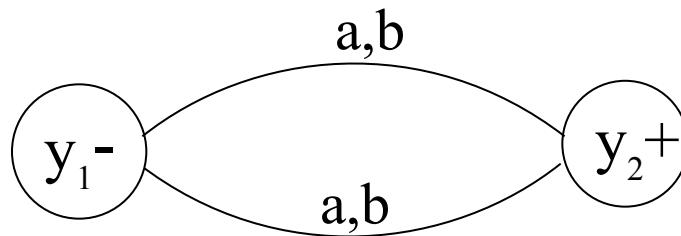


Old States	New States after reading	
	a	b
$z_1^- \sqcap (x_1, y_1)$	$(x_2, y_2) \sqcap z_2$	$(x_2, y_2) \sqcap z_2$

Example continued ...

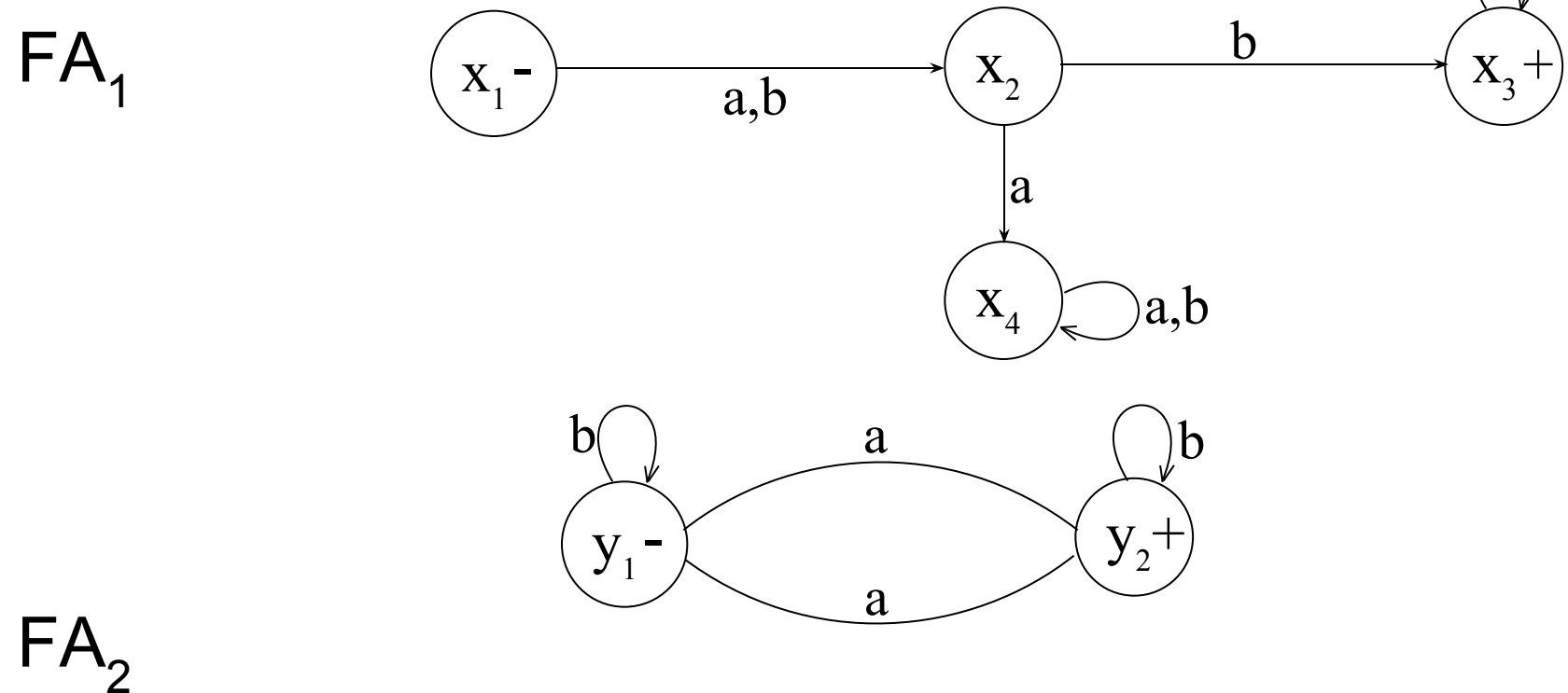
Old States	New States after reading	
	a	b
$z_2^+ \sqcap (x_2, y_2)$	$(x_1, y_1) \sqcap z_1$	$(x_1, y_1) \sqcap z_1$

Example continued ...



Task

Build FA corresponding to the concatenation of these two FAs *i.e.* FA_1FA_2 where



Kleene's Theorem Part III

Continued ...

- **Method3: (Closure of an FA)**

Building an FA corresponding to r^* , using the FA corresponding to r .

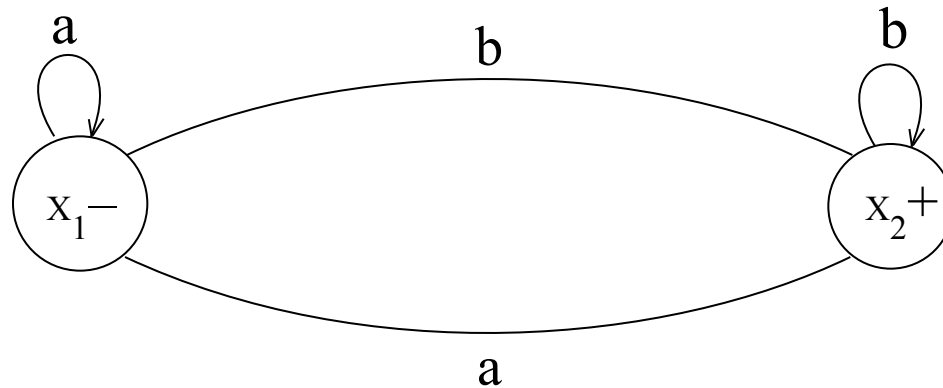
It is to be noted that if the given FA already accepts the language expressed by the closure of certain RE, then the given FA is the required FA. However the method, in other cases, can be developed considering the following examples

Closure of FA Continued ...

Closure of an FA, is same as concatenation of an FA with itself, except that the initial state of the required FA is a final state as well. Here the initial state of given FA, corresponds to the initial state of required FA and a non final state of the required FA as well.

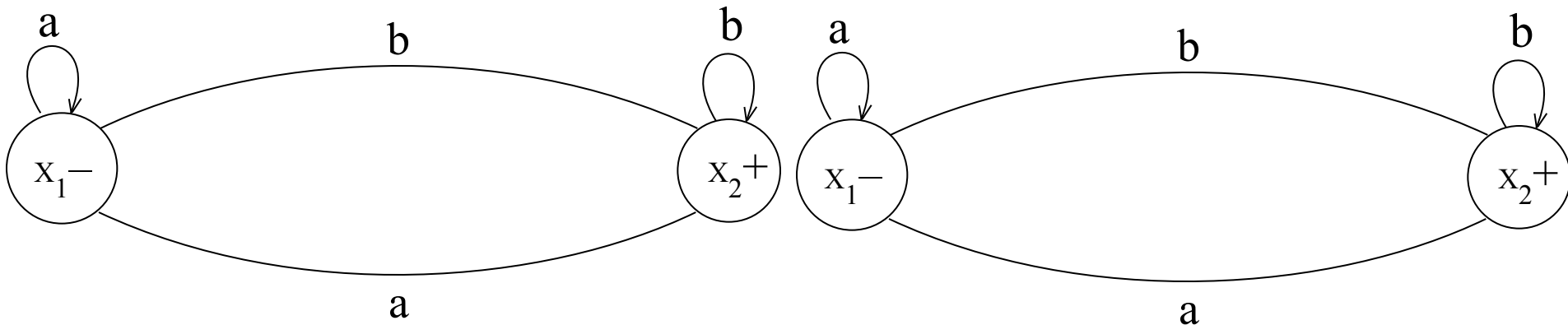
Example

Let $r=(a+b)^*b$ and the corresponding FA be



then the FA corresponding to r^* may be determined as under

Example continued ...



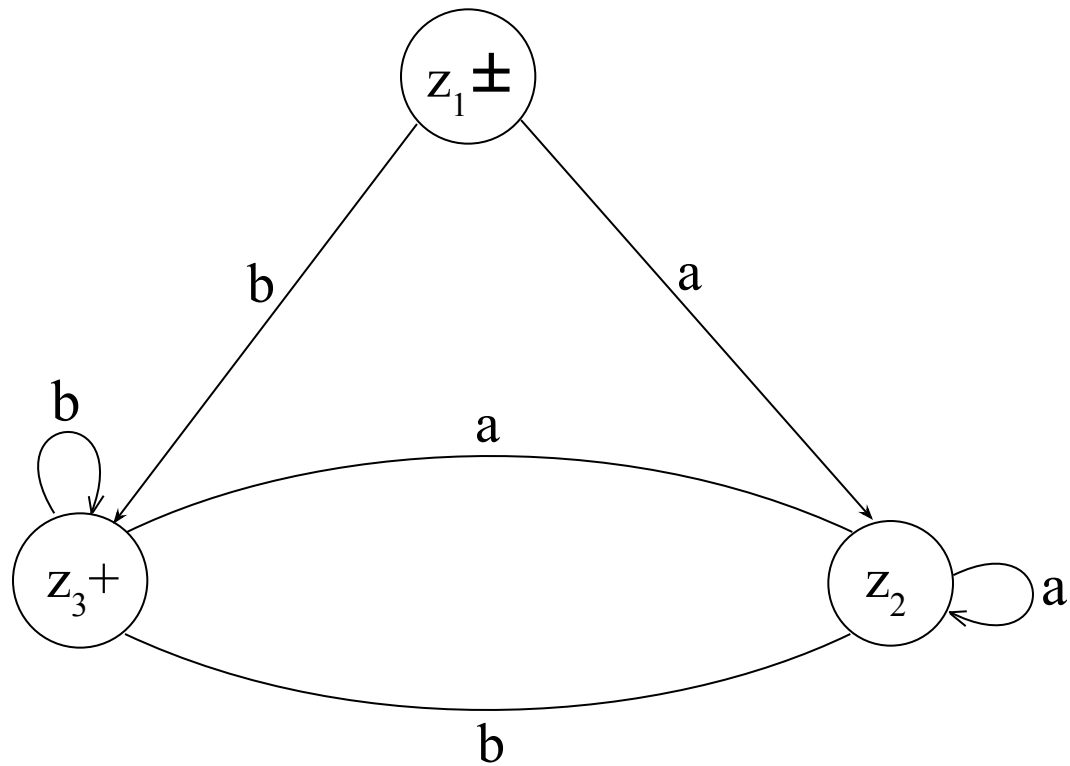
Old States	New States after reading	
	a	b
Final $z_1 \pm \square x_1$	$x_1 \square z_2$	$(x_2, x_1) \square z_{3_{17}}$

Example continued ...

Old States	New States after reading	
	a	b
Non-final $z_2 \sqcup x_1$	$x_1 \sqcup z_2$	$(x_2, x_1) \sqcup z_3$
$z_3^+ \sqcup (x_2, x_1)$	$x_1 \sqcup z_2$	$(x_2, x_1) \sqcup z_3$

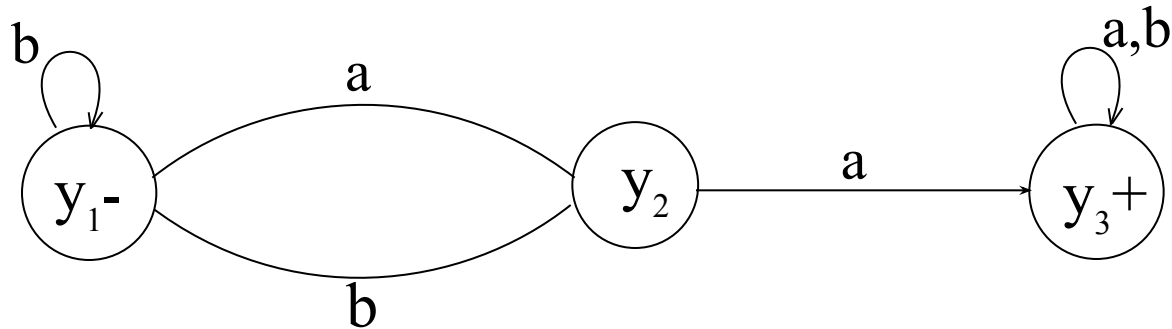
The corresponding transition diagram may be as under

Example continued ...

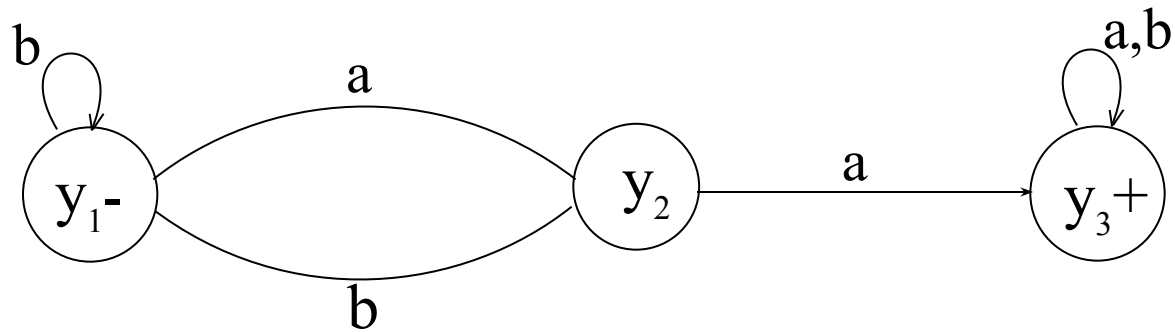
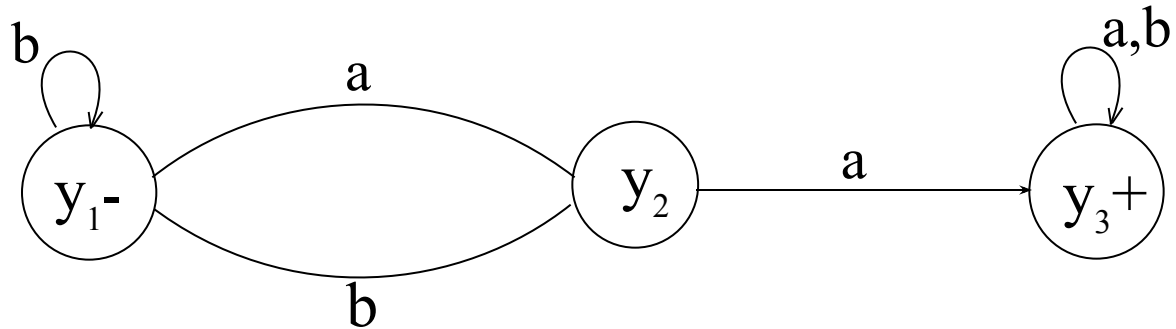


Example

Let $r=(a+b)^*aa(a+b)^*$ and the corresponding FA be



Example continued ...

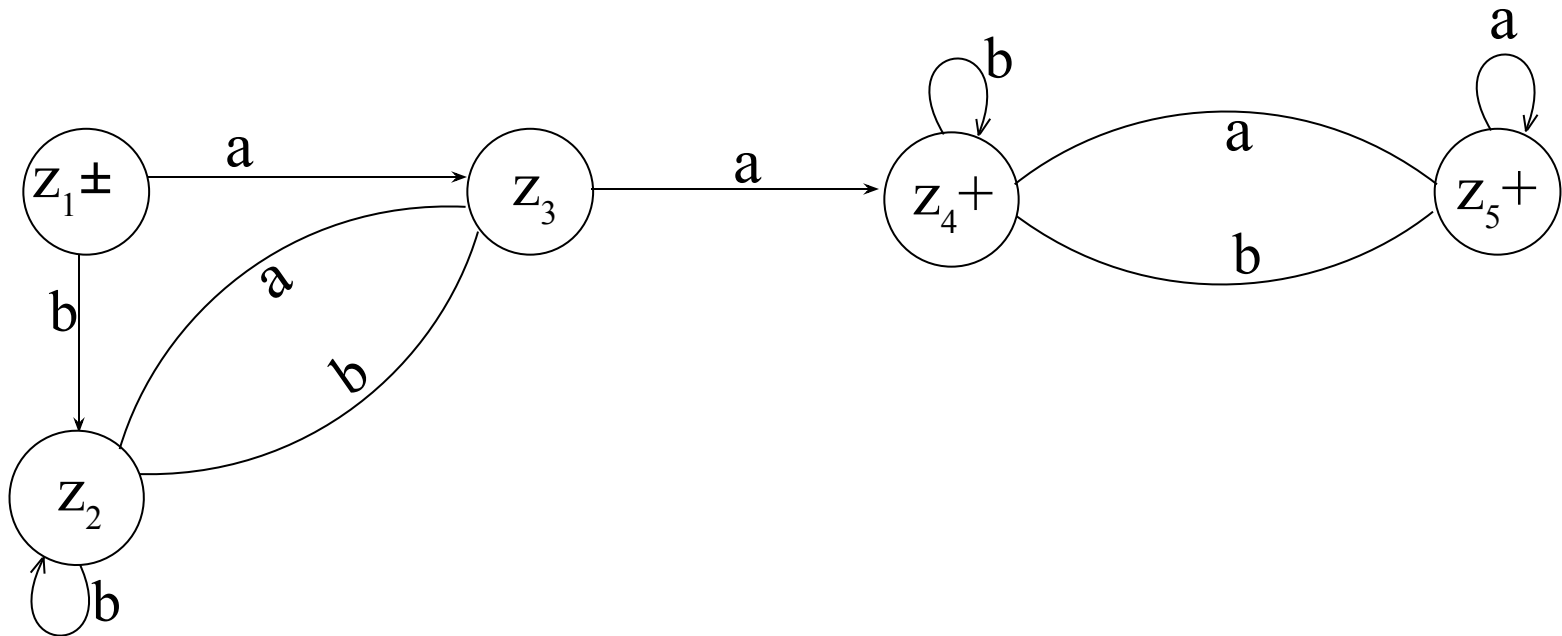


Old States	New States after reading	
	a	b
Final $z_1 \pm y_1$	$y_2 \square z_3$	$y_1 \square z_2$

Example continued ...

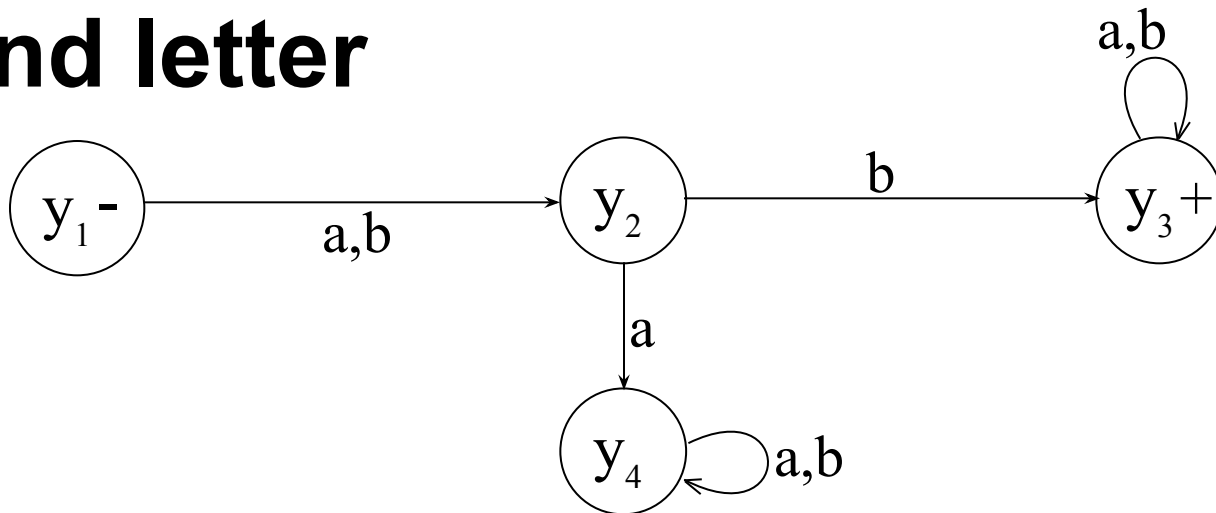
Old States	New States after reading	
	a	b
$z_2 \equiv y_1$	$y_2 \equiv z_3$	$y_1 \equiv z_2$
$z_3 \equiv y_2$	$(y_3, y_1) \equiv z_4$	$y_1 \equiv z_2$
$z_4^+ \equiv (y_3, y_1)$	$(y_3, y_1, y_2) \equiv z_5$	$(y_3, y_1) \equiv z_4$
$z_5^+ \equiv (y_3, y_1, y_2)$	$(y_3, y_1, y_2) \equiv z_5$	$(y_3, y_1) \equiv z_4$

Example continued ...

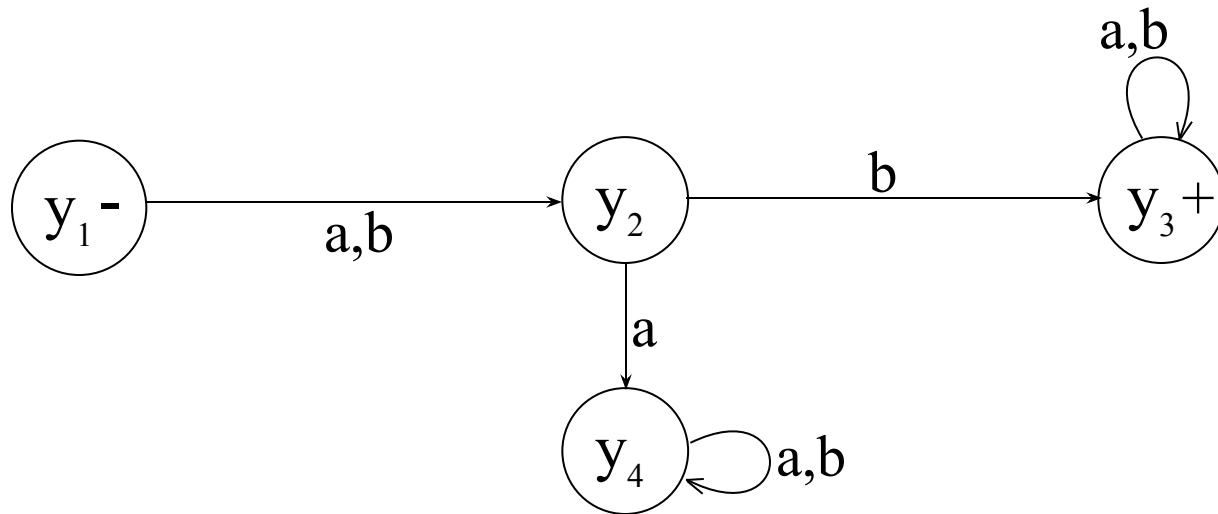


Example

Consider the following FA, accepting the language of strings with **b** as **second letter**



Example continued ...

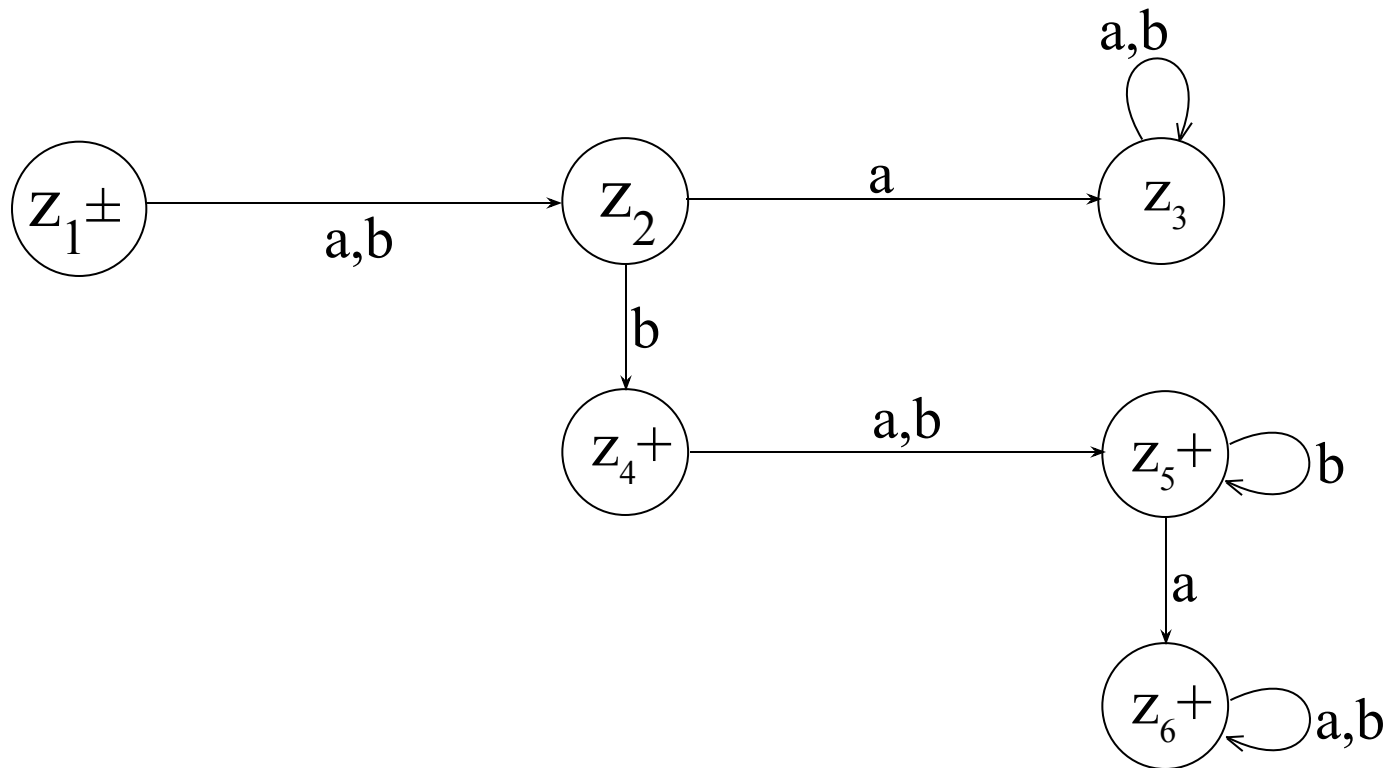


Old States	New States after reading	
	a	b
$z_1 \pm \square y_1$	$y_2 \square z_2$	$y_2 \square z_2$

Example continued ...

Old States	New States after reading	
	a	b
$z_2 \sqcup y_2$	$y_4 \sqcup z_3$	$(y_3, y_1) \sqcup z_4$
$z_3 \sqcup y_4$	$y_4 \sqcup z_3$	$y_4 \sqcup z_3$
$z_4^+ \sqcup (y_3, y_1)$	$(y_3, y_1, y_2) \sqcup z_5$	$(y_3, y_1, y_2) \sqcup z_5$
$z_5^+ \sqcup (y_3, y_1, y_2)$	$(y_3, y_1, y_2, y_4) \sqcup z_6$	$(y_3, y_1, y_2) \sqcup z_5$
$z_6 \sqcup (y_3, y_1, y_2, y_4)$	$(y_1, y_1, y_2, y_4) \sqcup z_6$	$(y_1, y_1, y_2, y_4) \sqcup z_6$

Example continued ...



Summing Up

- Examples of Kleene's theorem part III (method 1) continued, Kleene's theorem part III (method 2: Concatenation of FAs), Examples of Kleene's theorem part III (method 2: concatenation FAs) continued, Kleene's theorem part III (method 3: closure of an FA), examples of Kleene's theorem part III (method 3: Closure of an FA) continued